Restaurant Feedback Analyzer

Project Report

Name : Suyash Avatar
Reg. No. : 25BCE10367

GitHub Repository: https://github.com/suyashavatar/food-Feedback-Analyzer

1. Introduction

This project is a Natural Language Processing (NLP) based system designed to analyze restaurant reviews. It uses the VADER sentiment analyzer along with custom keyword-based enhancements to better understand how customers describe food, service, and experience.

2. Problem Statement

Restaurant platforms such as Zomato, Swiggy, Google Reviews, and TripAdvisor contain millions of customer reviews. Manually reading and analyzing these reviews is time-consuming and inefficient. This project provides an automated way to quickly extract sentiment insights from large volumes of text data.

3. Objectives

- Automate sentiment analysis of restaurant reviews
- Classify reviews into Positive, Negative, or Neutral
- Improve accuracy using domain-specific keyword adjustments
- Provide quick summary statistics

4. Course Concepts Applied

Concept	Application
Natural Language Processing	Processing and analyzing raw text
Sentiment Analysis	Classifying text into sentiment categories
VADER	Lexicon-based sentiment scoring
Keyword Enhancement	Improving accuracy using food-related words
Pandas	Handling CSV datasets

5. Methodology

The system works in two stages:

1. Base Sentiment Calculation using VADER

2. Adjustment using domain-specific keywords

Adjusted Score Formula:

adjusted_score = compound + (0.12 × positive_hits) - (0.12 × negative_hits) - (0.05 × neutral_hits)

Classification Thresholds:

>= 0.2 → Positive

<= -0.2 → Negative

Otherwise → Neutral

6. System Architecture

The project follows a modular structure:

main.py → Handles user interaction

analyzer.py → Contains sentiment logic

CSV dataset → Input data

7. Working

The system accepts input either through CSV files or manual sentence entry. Each review is processed using VADER to obtain a base score, which is then adjusted using predefined keywords. The final sentiment and score are displayed and optionally saved.

8. Sample Output

Review	Sentiment	Score
Best kofta in town, authentic taste	Positive	0.92
Very slow delivery; momos arrived cold	Negative	-0.24
The veg thali was okay, nothing special	Neutral	-0.19

9. Dataset Description

The dataset consists of 50 sample reviews from Indian restaurant platforms including Swiggy, Zomato, and Google Reviews. Each record contains review text, date, and source.

10. Advantages

- Fast and efficient analysis
- No training required (rule-based model)
- Easy to extend with new keywords

11. Limitations

- Limited to English language
- Cannot detect sarcasm accurately
- Depends on predefined keywords

12. Future Scope

- Add support for regional languages
- Integrate machine learning models
- Develop web interface (Flask/Streamlit)
- Add graphical visualization

13. Conclusion

This project demonstrates a practical implementation of sentiment analysis using NLP techniques. By combining VADER with domain-specific improvements, it provides meaningful insights into customer feedback.

14. References

- NLTK Documentation
- VADER Research Paper
- Online restaurant review platforms